

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 2: Basic Hydraulics (Fluids, Pumps & Accumulators)

Syllabus Source: CAE Oxford (pp. 47-98) | Pass Target: >= 90% AviationExam

1. FUNDAMENTAL HYDRAULIC PRINCIPLES & PASCAL'S LAW

Hydraulic systems transmit power through the medium of an incompressible liquid. Under Pascal's Law, pressure applied to an enclosed liquid is transmitted equally and undiminished in all directions, acting at right angles to the container walls:

- Formula for Pressure: Pressure (P) = Force (F) / Area (A). Units: PSI (Pounds per Square Inch) or Bar (1 bar = 14.5 psi). Standard commercial aircraft operate at 3,000 psi (e.g. A320, B737, B777) or 5,000 psi (A380, B787, A350).
- Mechanical Advantage: Force Out = Pressure x Area Out. By using a small input piston and a large actuator piston, massive output forces are generated, allowing effortless movement of primary flight controls under heavy aerodynamic loads.
- Work & Displacement: Work In = Work Out (Force x Distance). Liquid displaced by the small piston equals liquid moved into the large actuator: a small force moved over a long distance creates a large force moved over a short distance.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Hydraulic Force Multiplier & Accumulator

Formula: Force = Pressure x Piston Area. Example: Pressure 3,000 psi, piston area 5 sq in -> Output Force = 15,000 lbf! Accumulator Precharge Rule: - Gas chamber is precharged EXCLUSIVELY with DRY NITROGEN. - If precharge pressure is lost, the hydraulic system exhibits rapid needle fluctuations and loses emergency reserve braking.

2. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Pascal's Law and Hydraulic Force Multiplier

SCENARIO (Fundamental Aviation Hydraulic Calculation): A transport aircraft hydraulic system operates at 3,000 PSI (Pounds per Square Inch): - A flight control actuator cylinder has a piston face area A = 5 square inches (sq in) - The pilot commands an elevator deflection under high dynamic pressure cruise QUESTION: What is the total linear mechanical force generated by this hydraulic actuator on the flight control surface? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify Pascal's Master Equation: - Pressure = Force / Area --> Force = Pressure x Area Step 2: Plug in the Known Operating Values: - Hydraulic Operating Pressure (P) = 3,000 PSI - Actuator Working Area (A) = 5 sq in Step 3: Calculate the Resulting Output Force: - Force = 3,000 lbf/sq in x 5 sq in = 15,000 POUNDS OF FORCE (lbf)! - In metric conversion (1 lbf = 4.448 N): - Force = 15,000 x 4.448 N = 66,720 Newtons (~66.7 kN)! FINAL ANSWER: The actuator exerts 15,000 lbs (over 6.8 tonnes) of force on the elevator surface! This demonstrates how a compact 5-inch piston under 3,000 PSI effortlessly deflects massive flight surfaces against high-speed aerodynamic slipstream loads that no human pilot could move manually!

3. HYDRAULIC FLUIDS: MINERAL VS SYNTHETIC (SKYDROL)

Aviation hydraulic fluids must possess low viscosity change across extreme temperature ranges (-55°C to +130°C), high flash points, anti-foaming characteristics, and high lubricity. Two incompatible fluid types exist:

Parameter	Mineral Fluid (MIL-PRF-5606)	Synthetic Phosphate-Ester (Skydrol / HyJet)
Base Composition	Refined petroleum mineral base with chemical additives.	Synthetic phosphate-ester chemical base.
Color Code	DYED RED (to prevent servicing errors).	DYED PURPLE (or green in older Type IV formulations).
Flammability	Flammable! Flash point ~105°C; auto-ignition ~230°C.	Fire-resistant! Flash point > 175°C; auto-ignition > 475°C.
Seal Compatibility	Nitrile rubber (Buna-N), Neoprene.	Ethylene Propylene Diene Monomer (EPDM), Butyl, Teflon. DESTROYS natural rubber and Neoprene!
Toxicity & Hazards	Low toxicity; mild skin irritation.	HIGHLY TOXIC & CORROSIVE! Causes severe eye pain/stinging; damages aircraft acrylics and paint (requires polyurethane paint). Flush immediately with copious water!

[!] EXAM TRAP (AVIATIONEXAM) - Fluid Incompatibility & Seal Destruction

- NEVER MIX hydraulic fluids! If Skydrol is inadvertently serviced into a mineral system, the Nitrile/Neoprene seals swell, soften, and dissolve within minutes, causing total system failure. - If mineral oil is added to a Skydrol system, the EPDM seals rapidly degrade and fail. - Contaminated systems must be completely drained, flushed with chemical solvent, and have all seals replaced!

3. RESERVOIRS & PRESSURIZATION METHODS

The hydraulic reservoir stores fluid, accommodates thermal expansion, compensates for volume changes during jack extension/retraction, and permits air bubbles to separate out. In commercial aircraft, reservoirs MUST BE PRESSURIZED:

- Why Pressurize?: At high flight altitudes (FL 350+), low atmospheric ambient pressure would cause the hydraulic fluid to boil and cause cavitation (vapor pocket collapse) in the hydraulic pump suction inlet, leading to immediate pump destruction!
- Pressurization Sources: 1. Engine bleed air regulated to 30-50 psi; 2. Bootstrap reservoir (differential area piston utilizing system hydraulic pressure to pressurize the return fluid).

4. HYDRAULIC PUMPS: CONSTANT VS VARIABLE DISPLACEMENT

Pump Type	Operational Characteristics	Control Mechanism & Aircraft Use
Gear / Hand Pump	Constant volume displacement. Output flow is directly proportional to RPM.	Requires an Automatic Cut-Out Valve (ACOV) and system pressure relief valve to unseat and return fluid to reservoir when no actuators are moving.

Axial Piston Pump (EDP)	Variable volume displacement. Rotating cylinder block with angled swashplate (cam plate).	Swashplate angle modulates automatically via an internal pressure-compensating servo spool. At 3,000 psi demand satisfied, swashplate moves to 0° (zero delivery, zero power draw). Standard engine-driven pump (EDP).
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5. HYDRAULIC ACCUMULATORS & EMERGENCY POWER SOURCES

A hydraulic accumulator is a rigid pressure vessel divided into two chambers by a floating piston, flexible diaphragm, or bladder:

- **Gas Chamber:** Pre-charged with DRY NITROGEN (never compressed air or oxygen, which would cause an explosion in contact with hydraulic fluid mist!). Pre-charge is typically 1,000 to 1,500 psi.
- **Functions of Accumulator:** 1. Dampens pressure spikes and hydraulic hammering from rapid valve closure; 2. Supplements pump delivery during high transient flow demand (e.g. gear retraction); 3. Provides emergency pressure reserve to operate brakes or deploy gear following complete engine pump failure.
- **Power Transfer Unit (PTU):** A hydraulic motor driven by one system mechanically coupled to a pump in another system. Transfers HYDRAULIC POWER without any fluid transfer or intermixing!
- **Ram Air Turbine (RAT):** A deployable slipstream-driven air turbine that drives an emergency hydraulic pump, supplying essential flight controls in the event of dual engine flameout or total AC electrical failure.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why must hydraulic reservoirs in jet transport aircraft operating at high altitudes be pressurized?

- [A]: To prevent fluid freezing at -50°C
- [B]: To prevent hydraulic pump cavitation caused by low atmospheric ambient pressure at high altitudes
- [C]: To force fluid through the return filters
- [D]: To operate the landing gear emergency extension

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At high altitudes, low ambient pressure can cause the fluid to vaporize at the pump inlet. Pressurizing the reservoir (with engine bleed air to 30-50 psi) provides a positive head of pressure, preventing destructive cavitation.

Q2: What gas must be used to pre-charge a hydraulic accumulator, and why?

- [A]: Pure oxygen, to improve compressibility
- [B]: Compressed ambient air
- [C]: Dry nitrogen, because it is inert and will not support combustion or explode in contact with hydraulic oil
- [D]: Carbon dioxide

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Accumulators are pre-charged exclusively with dry nitrogen. Using compressed air or oxygen could cause diesel-effect auto-ignition or catastrophic explosion when mixed with hydraulic fluid mist.

Q3: What occurs if a mineral-based hydraulic fluid is accidentally serviced into an aircraft system using Skydrol?

- [A]: The fluid turns yellow but functions normally
- [B]: The seals (EPDM) in the Skydrol system swell, dissolve, and fail, causing rapid fluid loss and system destruction
- [C]: System operating pressure increases to 5,000 psi
- [D]: The fluid boils at 0°C

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Mineral oil and phosphate-ester (Skydrol) fluids are completely chemically incompatible. Adding mineral oil to a Skydrol system rapidly attacks and destroys the Ethylene Propylene seals.

Q4: What is the primary operational function of a Power Transfer Unit (PTU)?

- [A]: Transferring hydraulic fluid from a full system to a leaking system
- [B]: Transferring power between two independent hydraulic systems without transferring or mixing fluid
- [C]: Charging the nitrogen bottle
- [D]: Disconnecting the EDP during engine start

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A PTU consists of a hydraulic motor in one system connected via a mechanical shaft to a pump in another system. It allows power transfer without any fluid mixing.

Q5: In a variable displacement axial-piston hydraulic pump, what happens to the swashplate angle as system pressure increases?

- [A]: Swashplate angle increases to maximum
- [B]: Swashplate moves to approximately zero angle (perpendicular to piston axis), reducing pump output flow to zero
- [C]: Swashplate reverses rotation direction
- [D]: Swashplate disconnects from the driveshaft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. When system pressure satisfies the demand (3,000 psi), the pressure compensator moves the swashplate to near 0°, reducing piston stroke and output flow to zero, minimizing engine load.